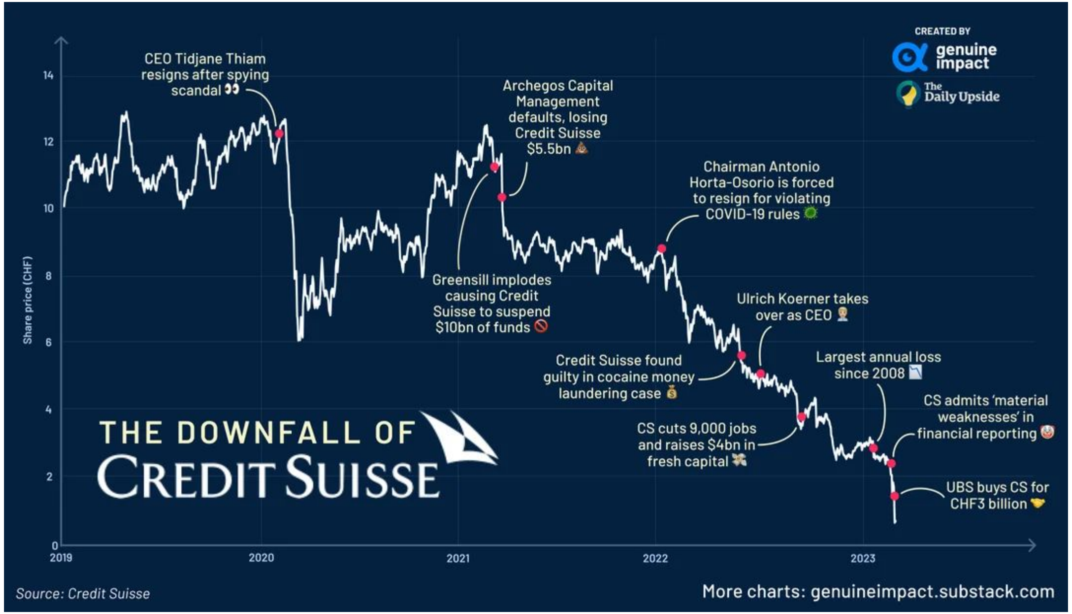

Merton's Credit Default Risk Model

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Motivation & Goal

- Credit Risk is backbone of the financial market
- The word "financing" refers to raising capital, one common approach is lending
    - US Treasuries: $28.3T (about 60% of total)
    - US corporate bonds: $11.2T (about 24% of that total)

- One natural question faced by lender is "Will the counterparty repay?"
- The question is non-trivial as one can see how poor credit risk management can harm the economy, e.g. the 2008 subprime brokerage financial crisis

- The goal of this project is to explore the framework of stochastic calculus in the application of credit risk market

Layout

1. Merton's Structural Default Risk Model (1974)
2. Credit Valuation Model
    a. Black-Scholes-Merton Model
    b. Heston Model
    c. Merton's Jump Diffusion Model (1976)
    d. Kou's Jump Diffusion Model (2002)
3. Application to Finance Discussion
4. Empirical Results Comparison & Evaluation
5. Appendix

Robert C. Merton

Corporate Default Definition

$$Asset_t(A_t) = Equity_t(E_t) + Debt_t(D_t)$$

$$E_t = A_t - D_t$$

$$\textbf{Default if } E_t \leq 0; \ \textbf{i.e.} \ A_t < D$$

Here we assume the debt is deterministic in time, Asset process follow a GBM

Merton's Credit Default Risk Model

Shareholder: $E_T = \max(A_T - D, 0)$

$$\frac{dA_t}{A_t} = \mu(A_t, t)dt + \sigma(A_t, t)dW(t)$$

Debt holder: $B_T = D - \max(D - A_T, 0)$

By FTAP,

$$E_t = c(S_t = A_t, K = D, T, \sigma, r)$$

$$D_t = D - p(S_t = A_t, K = D, T, \sigma, r)$$

Payoff are **long call** position and **short put** position for shareholder and debt holder respectively

Pricing of equity equal to pricing an **european call option**;

Pricing of debt equal to pricing a short position of an **european put option**

Pricing Model 1: Black-Scholes

$$\frac{dA_t}{A_t} = \mu dt + \sigma dW(t)$$

$$E_t = c^{BS}(S_t = A_t, K = D, T, \sigma, r)$$

$$D_t = D - p^{BS}(S_t = A_t, K = D, T, \sigma, r)$$

$$C(S_0, K, T, r, \sigma) = S_0\,\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

$$P(S_0, K, T, r, \sigma) = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1),$$

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

$$d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln(S_0/K) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

- Serves as the Benchmark

Pricing Model 2: Heston

$$dS_t = rS_t\,dt + \sqrt{v_t}\,S_t\,dW_t^{(1)},$$

$$dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^{(2)},$$

$$dW_t^{(1)}\,dW_t^{(2)} = \rho\,dt.$$

$$C_0 = S_0 P_1 - Ke^{-rT}P_2,$$

$$P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left(\frac{e^{-iu\ln K}\,\phi_j(u)}{iu}\right)du, \qquad j = 1, 2.$$

with

$$\phi_j(u) = \exp\Big(C_j(u,T) + D_j(u,T)\,v_0 + iu\ln S_0\Big),$$

$$d_j = \sqrt{(\rho\xi iu - b_j)^2 - \xi^2(2\alpha_j iu - u^2)},$$

$$g_j = \frac{b_j - \rho\xi iu + d_j}{b_j - \rho\xi iu - d_j},$$

$$C_j(u,T) = iurT + \frac{\kappa\theta}{\xi^2}\left[(b_j - \rho\xi iu + d_j)T - 2\ln\left(\frac{1 - g_j e^{d_j T}}{1 - g_j}\right)\right],$$

$$D_j(u,T) = \frac{b_j - \rho\xi iu + d_j}{\xi^2}\left(\frac{1 - e^{d_j T}}{1 - g_j e^{d_j T}}\right),$$

and parameter choices

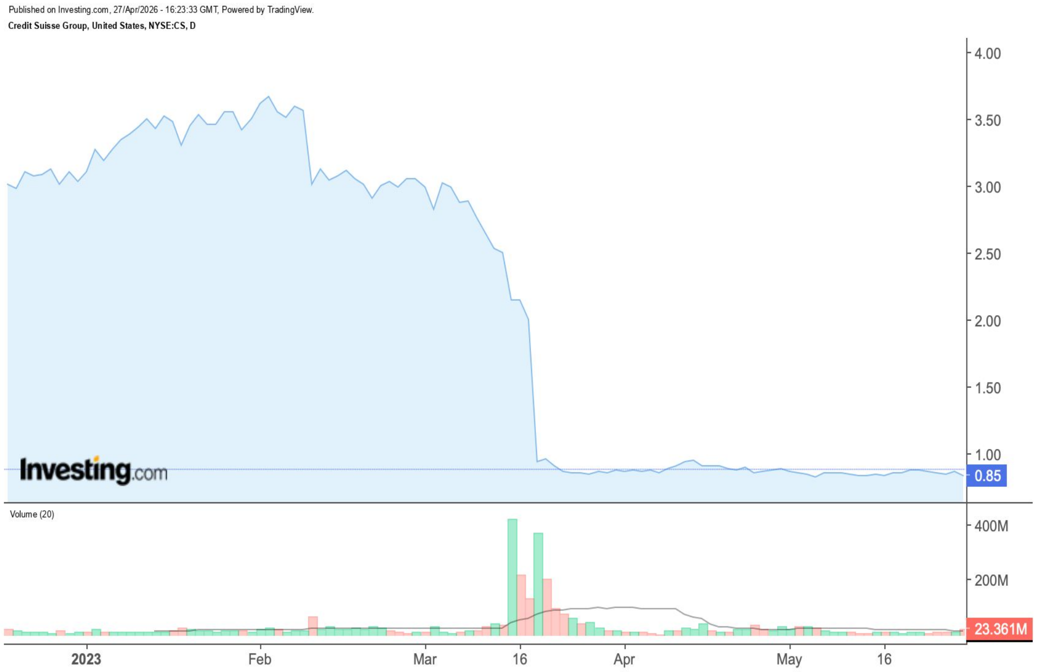
$$\alpha_1 = \frac{1}{2}, \quad \alpha_2 = -\frac{1}{2}, \qquad b_1 = \kappa - \rho\xi, \quad b_2 = \kappa.$$

Issue of Continuous Process Model

Diffusion Part: Captures small, continuous day-to-day fluctuations using Geometric Brownian Motion.

Jump Part: Accounts for sudden, large price movements using a Poisson process.

Risk Balance: Merton assumes jump risk is idiosyncratic and can be diversified away.



Poisson Process

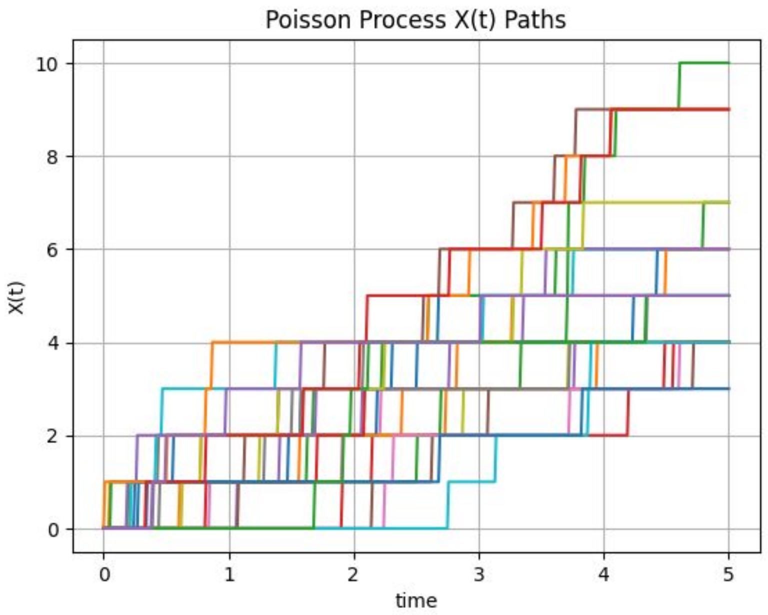
- **Definition.** A *(homogeneous) Poisson process* $(N_t)_{t\geq 0}$ with rate $\lambda > 0$ is a càdlàg counting process with $N_0 = 0$, independent increments, and

$$N_t - N_s \sim \text{Poisson}\big(\lambda(t-s)\big), \qquad 0 \leq s < t.$$

  Equivalently: jumps of size 1 at event times; interarrival times are i.i.d. $\text{Exp}(\lambda)$.

- **Finite-dimensional laws.** For disjoint intervals, counts are independent Poisson; superposition of independent Poisson processes is Poisson with summed rate.

- **Moments / transforms.** $E[N_t] = \lambda t$, $\text{Var}(N_t) = \lambda t$; pgf / mgf of N_t encodes the Poisson law.



Compound Compensated Poisson Process

- **Compensated Poisson process (martingale).**

$$M_t := N_t - \lambda t$$

is a martingale (w.r.t. the natural filtration, completed as usual). This is the discrete-jump analogue of Brownian "centered" part and is central in stochastic integration.

- **Compound Poisson process.** Let $(Y_n)_{n\geq 1}$ be i.i.d. real (or R^d) r.v.'s, independent of (N_t). The *compound Poisson process* is

$$X_t := \sum_{n=1}^{N_t} Y_n \quad (X_t := 0 \text{ if } N_t = 0).$$

- **Compensated compound Poisson process.** Assume additionally that

$$E[|Y_n|] < \infty.$$

The compound Poisson process

$$X_t := \sum_{n=1}^{N_t} Y_n$$

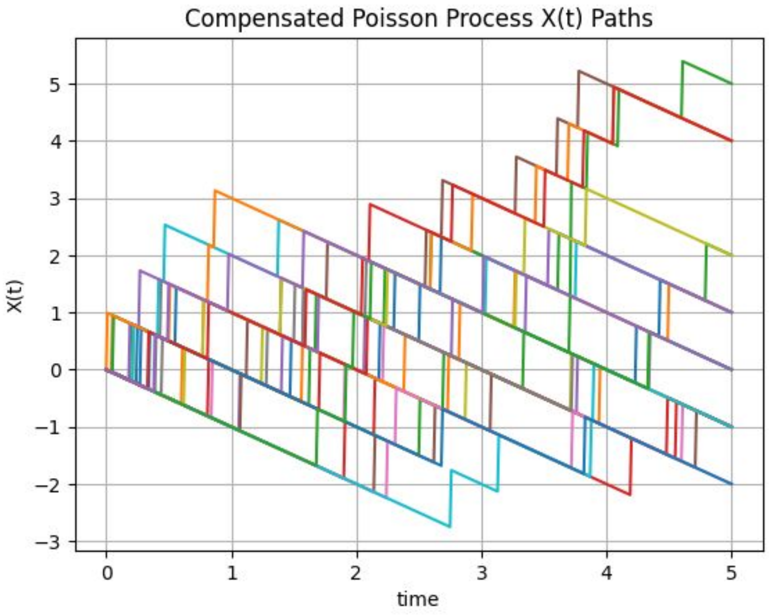
has expectation

$$E[X_t] = E[N_t]\, E[Y_n] = \lambda t\, E[Y_n].$$

Therefore its compensated version is

$$\widetilde{X}_t := X_t - \lambda t\, E[Y_n] = \sum_{n=1}^{N_t} Y_n - \lambda t\, E[Y_n].$$

Then $(\widetilde{X}_t)_{t\geq 0}$ is a martingale with respect to the natural filtration generated by (N_t) and the jump sizes (Y_n), completed as usual.



Model 3: Merton's Jump Diffusion Process

Assume the asset price A_t follows Merton's jump-diffusion process:

$$\frac{dA_t}{A_t} = (\mu - \lambda k)\, dt + \sigma\, dW(t) + d\left(\prod_{i=1}^{N_t} J_i - 1\right), \tag{1}$$

where:

- $W(t)$ is a standard Brownian motion,
- $N_t \sim \text{Poisson}(\lambda t)$ is the jump-counting process,
- $J_i > 0$ are i.i.d. jump multipliers with $k = E[J_i - 1]$,
- λk is the compensator drift correction, so A_t is a local martingale under the risk-neutral measure when $\mu = r$.

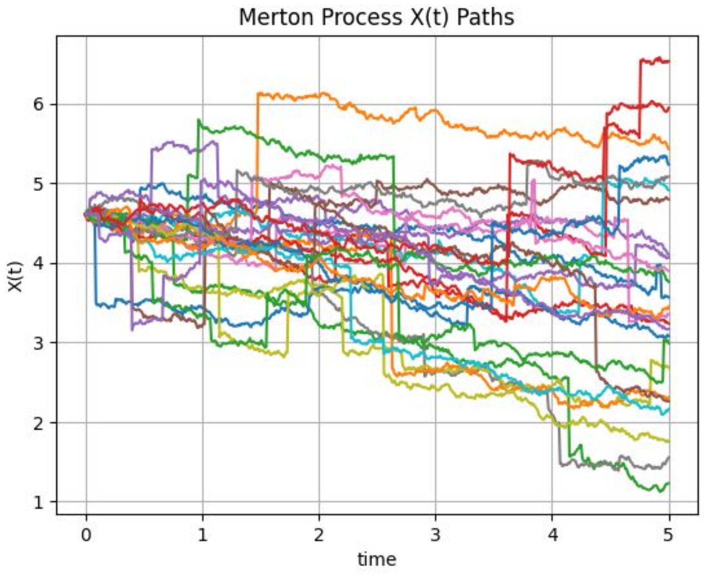
Write the dynamics more explicitly as

$$dA_t = (\mu - \lambda k)\, A_t\, dt + \sigma\, A_t\, dW(t) + A_{t^-}(J_{N_t} - 1)\, dN_t, \tag{2}$$

where A_{t^-} denotes the left-limit (pre-jump value) and dN_t marks each Poisson arrival. It can be rewritten as

$$\frac{dA_t}{A_t} = (\mu - \lambda k)dt + \sigma dW(t) + \sum_{n \in \mathbb{N}} (J_n - 1)\mathbf{1}(\Gamma_n = t) \tag{3}$$

where $\Gamma(n) = \sum_{i=1}^{n} Z_i$ denotes the arrival time of n-th arrival, Z_i is defined to be exponentially distributed.



Itô's Lemma for Jump Diffusion Process

Applying Itô's Lemma for Jump-Diffusion Processes

Let $f(A) = \log A$. For a jump-diffusion process the generalised Itô formula is

$$df(A_t) = f'(A_t)\,dA_t^c + \tfrac{1}{2}f''(A_t)(dA_t^c)^2 + \left[f(A_{t^-}J_{N_t}) - f(A_{t^-})\right]dN_t, \tag{4}$$

where $dA_t^c = (\mu - \lambda k)A_t\,dt + \sigma A_t\,dW(t)$ is the continuous part.

Computing each term with $f'(A) = 1/A$ and $f''(A) = -1/A^2$:

$$f'(A_t)\,dA_t^c = \frac{1}{A_t}\left[(\mu - \lambda k)A_t\,dt + \sigma A_t\,dW(t)\right] = (\mu - \lambda k)\,dt + \sigma\,dW(t), \tag{5}$$

$$\tfrac{1}{2}f''(A_t)(dA_t^c)^2 = \tfrac{1}{2}\left(-\frac{1}{A_t^2}\right)\sigma^2 A_t^2\,dt = -\tfrac{1}{2}\sigma^2\,dt, \tag{6}$$

$$f(A_{t^-}J_{N_t}) - f(A_{t^-}) = \log(A_{t^-}J_{N_t}) - \log(A_{t^-}) = \log J_{N_t}. \tag{7}$$

Substituting back:

$$d\log A_t = \left(\mu - \lambda k - \tfrac{1}{2}\sigma^2\right)dt + \sigma\,dW(t) + \log J_{N_t}\,dN_t. \tag{8}$$

Asset Density under Merton's Model

Integrating to Obtain $\log(A_T/A_0)$

Integrating from 0 to T and summing the jump contributions:

$$\log \frac{A_T}{A_0} = \left(\mu - \lambda k - \tfrac{1}{2}\sigma^2\right) T + \sigma W(T) + \sum_{i=1}^{N_T} \log J_i, \tag{9}$$

so that

$$\boxed{A_T = A_0 \exp\left[\left(\mu - \lambda k - \tfrac{1}{2}\sigma^2\right) T + \sigma W(T)\right] \prod_{i=1}^{N_T} J_i.} \tag{10}$$

Under the risk-neutral measure, set $\mu = r$ throughout (and keep λk as the compensator):

$$A_T = A_0 \exp\left[\left(r - \lambda k - \tfrac{1}{2}\sigma^2\right) T + \sigma W(T)\right] \prod_{i=1}^{N_T} J_i. \tag{11}$$

Lognormal Jumps (Merton's Specification)

Following Merton (1976), assume $\log J_i \sim \mathcal{N}(\gamma, \delta^2)$ so that

$$k = E[J_i - 1] = e^{\gamma + \frac{1}{2}\delta^2} - 1, \tag{12}$$

$$E[\log(1+\varepsilon_i)] = \gamma - \tfrac{1}{2}\delta^2, \qquad \mathrm{Var}[\log(1+\varepsilon_i)] = \delta^2, \tag{13}$$

where $\varepsilon_i = J_i - 1$.

Conditioning on $N_T = n$

Conditional on $N_T = n$, the sum $\sum_{i=1}^n \log J_i \sim \mathcal{N}(n\gamma,\ n\delta^2)$. Therefore, conditional on $N_T = n$, the log-return is normally distributed:

$$\log \frac{A_T}{A_0} \,\Big|\, N_T = n \sim \mathcal{N}\left(\left(r - \lambda k - \tfrac{1}{2}\sigma^2\right) T + n\gamma,\ \sigma^2 T + n\delta^2\right). \tag{14}$$

This means $A_T \mid N_T = n$ is lognormally distributed, exactly as required by the Black–Scholes model, but with adjusted parameters:

$$\sigma_n^2 = \sigma^2 + \frac{n\delta^2}{T}, \qquad r_n = r - \lambda k + \frac{n\gamma}{T}. \tag{15}$$

Conditional distribution given $N_T = n$. From

$$\log\left(\frac{A_T}{A_0}\right) = \left(r - \lambda k - \frac{1}{2}\sigma^2\right) T + \sigma W_T + \sum_{i=1}^{N_T} \log J_i,$$

and Merton's assumption $\log J_i \sim \mathcal{N}(\gamma, \delta^2)$, we have

$$\sum_{i=1}^{n} \log J_i \sim \mathcal{N}(n\gamma,\ n\delta^2).$$

Hence, conditional on $N_T = n$,

$$\log\left(\frac{A_T}{A_0}\right) \Big|\, N_T = n \sim \mathcal{N}\left(\left(r - \lambda k - \frac{1}{2}\sigma^2\right) T + n\gamma,\ \sigma^2 T + n\delta^2\right).$$

Define adjusted Black–Scholes parameters

$$\sigma_n^2 := \sigma^2 + \frac{n\delta^2}{T}, \qquad r_n := r - \lambda k + \frac{n\gamma}{T}.$$

Then equivalently

$$\log\left(\frac{A_T}{A_0}\right) \Big|\, N_T = n \sim \mathcal{N}\left(\left(r_n - \frac{1}{2}\sigma_n^2\right) T,\ \sigma_n^2 T\right),$$

so

$$A_T \mid N_T = n \sim \mathrm{Lognormal}\left(\log A_0 + \left(r_n - \frac{1}{2}\sigma_n^2\right) T,\ \sigma_n^2 T\right).$$

Therefore, conditional on $N_T = n$, A_T has the same form as in Black–Scholes with parameters (r_n, σ_n).

Option Pricing under Merton's Model

European Call Price as an Infinite Sum of Black–Scholes Prices

The risk-neutral price of a European call with strike D and maturity T is

$$C^{\mathrm{JD}} = e^{-rT}\, E^{\mathbb{Q}}[\max(A_T - D,\, 0)]\,. \tag{19}$$

Apply the tower property, conditioning on N_T:

$$C^{\mathrm{JD}} = e^{-rT} \sum_{n=0}^{\infty} P(N_T = n)\; E^{\mathbb{Q}}\big[\max(A_T - D,\, 0) \,\big|\, N_T = n\big]\,. \tag{20}$$

Because $A_T \mid N_T = n$ is lognormal with parameters (σ_n^2, r_n), the inner expectation is exactly the undiscounted Black–Scholes call payoff with those parameters. Absorbing the e^{-rT} discount into each BS term:

$$E^{\mathbb{Q}}\big[\max(A_T - D, 0) \,\big|\, N_T = n\big] \cdot e^{-rT} = e^{-(r_n - r_n + r)T}\,(\ldots) = C^{\mathrm{BS}}(A_t,\, D,\, T,\, \sigma_n^2,\, r_n), \tag{21}$$

where C^{BS} is the standard Black–Scholes call price:

$$C^{\mathrm{BS}}(A_t, D, T, \sigma_n^2, r_n) = A_t\, \Phi(d_1) - D\, e^{-r_n T}\, \Phi(d_2), \tag{22}$$

$$d_1 = \frac{\log(A_t/D) + (r_n + \frac{1}{2}\sigma_n^2)\, T}{\sigma_n \sqrt{T}}, \qquad d_2 = d_1 - \sigma_n \sqrt{T}. \tag{23}$$

$$\boxed{C^{\mathrm{JD}}(A_t, D, T, \sigma^2, r, \delta^2, \lambda, k) = \sum_{n=0}^{\infty} \frac{(\lambda' T)^n}{n!}\, e^{-\lambda' T}\, C^{\mathrm{BS}}(A_t,\, D,\, T,\, \sigma_n^2,\, r_n),} \tag{26}$$

with

$$\lambda' = \lambda(1 + k), \tag{27}$$

$$r_n = r - \lambda k + \frac{n\gamma}{T}, \tag{28}$$

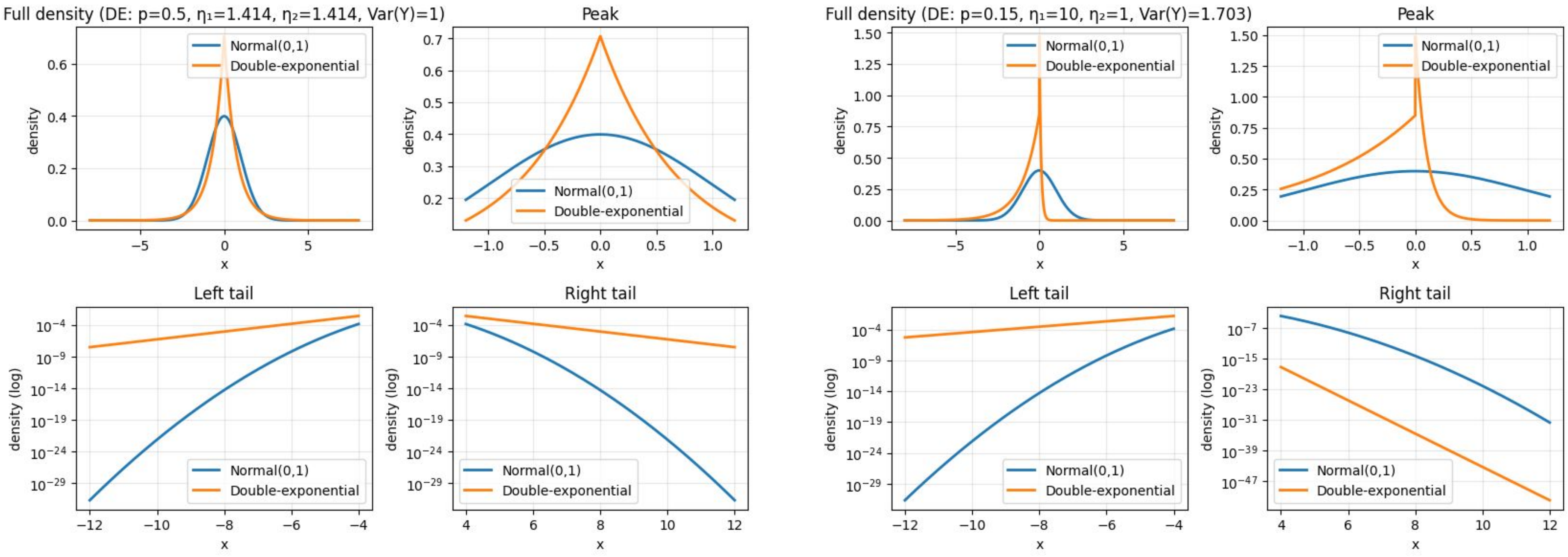
$$\sigma_n^2 = \sigma^2 + \frac{n\delta^2}{T}, \tag{29}$$

$$\gamma = \log(1 + k). \tag{30}$$

Takeaway Each term in the sum is the Black–Scholes price computed as if exactly n jumps occurred, weighted by the Poisson probability of observing exactly n jumps over $[0, T]$. The adjusted drift r_n accounts for the mean jump contribution and the compensator, while σ_n^2 inflates the instantaneous variance to reflect the additional randomness from the n jump sizes.

Double Exponential vs Gaussian

1) Leverage Effect 2) leptokurtosis



Model 4: Kou's Jump Diffusion Process

Asset Price Dynamics

Assume the asset price P_t follows Kou's jump-diffusion process under the physical measure:

$$\frac{dP_t}{P_t} = \mu\, dt + \sigma\, dW_t + d\left(\sum_{i=1}^{n_t}(J_i - 1)\right), \tag{1}$$

where:

- W_t is a standard Wiener process (Brownian motion),
- $n_t \sim \text{Poisson}(\lambda t)$ is the jump-counting process with rate λ,
- $J_i \sim \mathit{Laplace}(p, \eta_1, \eta_2)$ are i.i.d. jump multipliers, with $k = E[J_i - 1]$,
- n_t, W_t, and $\{J_i\}$ are independent.

Double Exponential Distribution (Laplacian Distribution)

The key innovation in Kou's model is that $X = \log(J)$ has an **asymmetric double exponential distribution** (also called the Laplacian distribution) with probability density function:

$$f_X(x) = p\,\eta_1\, e^{-\eta_1 x}\, \mathbf{1}_{\{x\geq 0\}} + q\,\eta_2\, e^{\eta_2 x}\, \mathbf{1}_{\{x<0\}}, \quad 0 < \eta_1, \eta_2 < \infty, \tag{2}$$

where $p, q \geq 0$, $p + q = 1$, and $\eta_1 > 1$, $\eta_2 > 0$.

Alternative representation: The double exponential distribution can be written as

$$X - \kappa = \begin{cases} \xi & \text{with probability } p, \\ -\xi & \text{with probability } q, \end{cases} \tag{3}$$

where $\xi \sim \mathit{Expo}(1/\eta)$ with mean η and variance η^2, with density

$$f(\xi) = \frac{1}{\eta} e^{-\xi/\eta}, \quad 0 < \xi < \infty. \tag{4}$$

Equivalently, we can write:

$$\log(J_i) = X = \begin{cases} \xi_+ & \text{with probability } p, \\ -\xi_- & \text{with probability } q, \end{cases} \tag{5}$$

where ξ_+ and ξ_- are exponential random variables with means $1/\eta_1$ and $1/\eta_2$, respectively.

Kou's Asset Process & Simple Return

Asset Price

Solving the stochastic differential equation yields the dynamics of the asset price:

$$A_t = A_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right] \prod_{i=1}^{n_t} J_i. \tag{9}$$

Note that $\prod_{i=1}^{0} J_i = 1$ by convention (when there are no jumps).

Return Distribution over Small Time Increments

From the asset price dynamics, the simple return over a time interval Δt is:

$$\frac{A_{t+\Delta t} - A_t}{A_t} = \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma(W_{t+\Delta t} - W_t) + \sum_{i=n_t+1}^{n_{t+\Delta t}} X_i\right] - 1, \tag{10}$$

where $X_i = \log(J_i)$ and the summation over an empty set is zero. For small Δt, using the approximation $e^x \approx 1 + x + x^2/2$ and $(\Delta W_t)^2 \approx \Delta t$:

$$\frac{A_{t+\Delta t} - A_t}{A_t} \approx \left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\,\Delta W_t + \sum_{i=n_t+1}^{n_{t+\Delta t}} X_i + \frac{1}{2}\sigma^2(\Delta W_t)^2 \tag{11}$$

$$\approx \mu\,\Delta t + \sigma\,\epsilon\sqrt{\Delta t} + \sum_{i=n_t+1}^{n_{t+\Delta t}} X_i, \tag{12}$$

where $\Delta W_t = W_{t+\Delta t} - W_t$ and ϵ is a standard normal random variable.

Combined Return Distribution

Combining the prior results, the simple return of the underlying asset is approximately distributed as:

$$\frac{A_{t+\Delta t} - A_t}{A_t} \approx \mu\,\Delta t + \sigma\,\epsilon\sqrt{\Delta t} + I \times X, \tag{14}$$

where:

- I is a Bernoulli random variable with $\Pr(I = 1) = \lambda\,\Delta t$ and $\Pr(I = 0) = 1 - \lambda\,\Delta t$,
- X is a double exponential random variable defined in equation (5),
- I, ϵ, and X are independent.

Equation (10) reduces to geometric Brownian motion without jumps when $\lambda = 0$.

Density Function of the Return

Let $G = \mu\,\Delta t + \sigma\,\epsilon\sqrt{\Delta t} + I \times X$ be the random variable on the right-hand side of equation (10). Using the independence between the exponential and normal distributions, Kou (2002) obtains the probability density function of G as:

$$g(x) = \frac{\lambda\,\Delta t}{2\eta} e^{\sigma^2\Delta t/(2\eta^2)} \left[e^{-\omega/\eta}\,\Phi\left(\frac{\omega\eta - \sigma^2\Delta t}{\sigma\eta\sqrt{\Delta t}}\right) + e^{\omega/\eta}\,\Phi\left(\frac{\omega\eta + \sigma^2\Delta t}{\sigma\eta\sqrt{\Delta t}}\right)\right]$$
$$+ (1 - \lambda\,\Delta t)\,\frac{1}{\sigma\sqrt{\Delta t}}\,\phi\left(\frac{x - \mu\,\Delta t}{\sigma\sqrt{\Delta t}}\right), \tag{15}$$

where $\omega = x - \mu\,\Delta t - \kappa$, and $\phi(\cdot)$ and $\Phi(\cdot)$ are the probability density and cumulative distribution functions of the standard normal random variable, respectively.

Asset Density under Kou's Model

Asset price under Kou's double-exponential jump-diffusion.

Under the physical measure, Kou specifies

$$\frac{dA_t}{A_{t-}} = \mu\,dt + \sigma\,dW_t + d\left(\sum_{i=1}^{N_t}(J_i - 1)\right), \tag{18}$$

with $N_t \sim \text{Poisson}(\lambda t)$ independent of W, and i.i.d. jump sizes such that $Y_i := \log J_i$ has the asymmetric double-exponential density

$$f_Y(y) = p\,\eta_1 e^{-\eta_1 y}\mathbf{1}_{\{y\geq 0\}} + q\,\eta_2 e^{\eta_2 y}\mathbf{1}_{\{y<0\}}, \qquad \eta_1 > 1,\ \eta_2 > 0,\ \ p + q = 1. \tag{19}$$

Integrating the dynamics gives the closed-form level (Kou's equation for S_t):

$$A_t = A_0\,\exp\!\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)T + \sigma W_T\right]\prod_{i=1}^{N_T} J_i = A_0\,\exp\!\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)T + \sigma W_T + \sum_{i=1}^{N_T} Y_i\right], \tag{20}$$

with the convention $\prod_{i=1}^{0}(\cdot) = 1$.

Conditional distribution given $N_T = n$**.** Fix $n \in \{0, 1, 2, \ldots\}$. Conditional on $N_T = n$, the Wiener increment W_T and the n jump logs $\{Y_i\}_{i=1}^{n}$ are independent, hence

$$\log\frac{A_t}{A_0}\,\Big|\,(N_T = n) \;=\; \left(\mu - \tfrac{1}{2}\sigma^2\right)T + \sigma W_T + \sum_{i=1}^{n} Y_i. \tag{21}$$

Equivalently, writing $Z \sim \mathcal{N}(0, 1)$,

$$\log\frac{A_t}{A_0}\,\Big|\,(N_T = n) \overset{d}{=} \left(\mu - \tfrac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z + \sum_{i=1}^{n} Y_i. \tag{22}$$

Relation to the Merton (lognormal-jump) case.

- If $n = 0$, then $\sum_{i=1}^{0} Y_i = 0$ and

$$\log\frac{A_t}{A_0}\,\Big|\,(N_T = 0) \sim \mathcal{N}\!\left(\left(\mu - \tfrac{1}{2}\sigma^2\right)T,\ \sigma^2 T\right),$$

  so $A_t \mid (N_T = 0)$ is *exactly lognormal* (pure diffusion).
- If $n \geq 1$, the jump contribution $\sum_{i=1}^{n} Y_i$ is *not* Gaussian under the double-exponential law for Y, so in general $\log(A_t/A_0) \mid (N_T = n)$ is *not* normal and therefore $A_t \mid (N_T = n)$ is *not* lognormal (unlike Merton's Gaussian log-jumps).

Marginal (unconditional) distribution

Unconditionally,

$$A_t = \underbrace{\sum_{n=0}^{\infty}\frac{(\lambda T)^n}{n!}e^{-\lambda T}}_{\Pr(N_T = n)}\cdot\mathbb{E}\!\left[A_t \mid N_T = n\right] \quad \text{(tower property for expectations)}, \tag{23}$$

and more generally for any payoff $\varphi(\cdot)$,

$$\mathbb{E}[\varphi(A_t)] = \sum_{n=0}^{\infty}\frac{(\lambda T)^n}{n!}e^{-\lambda T}\,\mathbb{E}[\varphi(A_t) \mid N_T = n]. \tag{24}$$

The conditional expectation $\mathbb{E}[\,\cdot \mid N_T = n]$ involves the convolution of a normal with the sum of n i.i.d. double exponentials (this is where Kou's tractability via exponential-type structure enters), but the conditional *law* of A_t does not collapse to a single lognormal family indexed by n the way it does in Merton.

Option Pricing under Kou’s Model

1 Option Pricing Under Kou’s Jump-Diffusion Model

1.1 Risk-Neutral Dynamics

In the presence of random jumps, the market becomes incomplete and standard hedging arguments are not applicable. However, assuming jump risk is diversifiable, we can derive an option pricing formula under a risk-neutralmeasure $\mathbf{Q}$ such that

$$\begin{aligned}\frac{dA_t}{A_t} &= [r - \lambda E(J-1)]\,dt + \sigma\,dW_t + d\left(\sum_{i=1}^{n_t}(J_i - 1)\right) \\ &= (r - \lambda\psi)\,dt + \sigma\,dW_t + d\left(\sum_{i=1}^{n_t}(J_i - 1)\right),\end{aligned} \tag{26}$$

where r is the risk-free interest rate, $J = e^X$ such that X follows the double exponential distribution with density f_X given in Eq.,

$$\psi = E[J-1] = E[e^X] - 1 = \frac{e^\kappa}{1-\eta^2} - 1, \quad 0 < \eta < 1, \tag{27}$$

and the parameters κ, η, ψ, σ become risk-neutral parameters taking consideration of risk premiums.

1.2 Risk-Neutral Asset Price

The unique solution of the risk-neutral dynamics is

$$A_t = A_0 \exp\left[\left(r - \frac{\sigma^2}{2} - \lambda\psi\right)t + \sigma W_t\right]\prod_{i=1}^{n_t} J_i. \tag{28}$$

1.3 European Call Option Price

The price of a European call option at time t is given by

$$c_t = E_*\left[e^{-r(T-t)}(A_t - K)_+\right], \tag{29}$$

where $(y)_+ = \max(0, y)$, K is the strike price, T is the expiration time, and the expectation is taken under the risk-neutral measure $\mathbf{Q}$. Explicitly:

$$c_t = E^{\mathbf{Q}}\left[e^{-r(T-t)}\left(A_t \exp\left[\left(r - \frac{\sigma^2}{2} - \lambda\psi\right)(T-t) + \sigma\sqrt{T-t}\,\epsilon\right]\prod_{i=1}^{n_T} J_i - K\right)_+\right], \tag{30}$$

where ϵ is a standard normal random variable.

Option Pricing under Kou's Model

1.4 Analytical Pricing Formula

Kou (2002) shows that c_t is analytically tractable as

$$c_t = \sum_{n=1}^{\infty}\sum_{j=1}^{n} e^{-\lambda(T-t)}\frac{\lambda^n (T-t)^n}{n!}\frac{2^j}{2^{2n-1}}\binom{2n-j-1}{n-1}\left(A_{1,n,j}+A_{2,n,j}+A_{3,n,j}\right)$$
$$+ e^{-\lambda(T-t)}\left[A_t\, e^{-\lambda\psi(T-t)}\Phi(h_+) - Ke^{-r(T-t)}\Phi(h_-)\right], \tag{31}$$

where $\Phi(\cdot)$ is the CDF of the standard normal random variable and

$$A_{1,n,j} = A_t\, e^{-\lambda\psi(T-t)+n\kappa}\frac{1}{2}\left(\frac{1}{(1-\eta)^j}+\frac{1}{(1+\eta)^j}\right)\Phi(b_+) - e^{-r(T-t)}K\,\Phi(b_-), \tag{32}$$

$$A_{2,n,j} = \frac{1}{2}\,e^{-r(T-t)-\omega/\eta+\sigma^2(T-t)/(2\eta^2)}\,K\sum_{i=0}^{j-1}\left(\frac{1}{(1-\eta)^{j-i}}-1\right)\left(\frac{\sigma\sqrt{T-t}}{\eta}\right)^i\frac{1}{\sqrt{2\pi}}Hh_i(c_-), \tag{33}$$

$$A_{3,n,j} = \frac{1}{2}\,e^{-r(T-t)+\omega/\eta+\sigma^2(T-t)/(2\eta^2)}\,K\sum_{i=0}^{j-1}\left(1-\frac{1}{(1+\eta)^{j-i}}\right)\left(\frac{\sigma\sqrt{T-t}}{\eta}\right)^i\frac{1}{\sqrt{2\pi}}Hh_i(c_+), \tag{34}$$

with auxiliary quantities

$$b_\pm = \frac{\ln(A_t/K)+(r\pm\sigma^2/2-\lambda\psi)(T-t)+n\kappa}{\sigma\sqrt{T-t}}, \tag{35}$$

$$h_\pm = \frac{\ln(A_t/K)+(r\pm\sigma^2/2-\lambda\psi)(T-t)}{\sigma\sqrt{T-t}}, \tag{36}$$

$$c_\pm = \frac{\sigma\sqrt{T-t}}{\eta} \pm \frac{\omega}{\sigma\sqrt{T-t}}, \tag{37}$$

$$\omega = \ln(K/A_t)+\lambda\psi(T-t)-\left(r-\sigma^2/2\right)(T-t)-n\kappa, \tag{38}$$

$$\psi = \frac{e^\kappa}{1-\eta^2}-1. \tag{39}$$

1.5 The Hh Functions

The $Hh_n(\cdot)$ functions are defined by

$$Hh_n(x) = \frac{1}{n!}\int_x^\infty (s-x)^n e^{-s^2/2}\,ds, \quad n=0,1,2,\ldots, \tag{40}$$

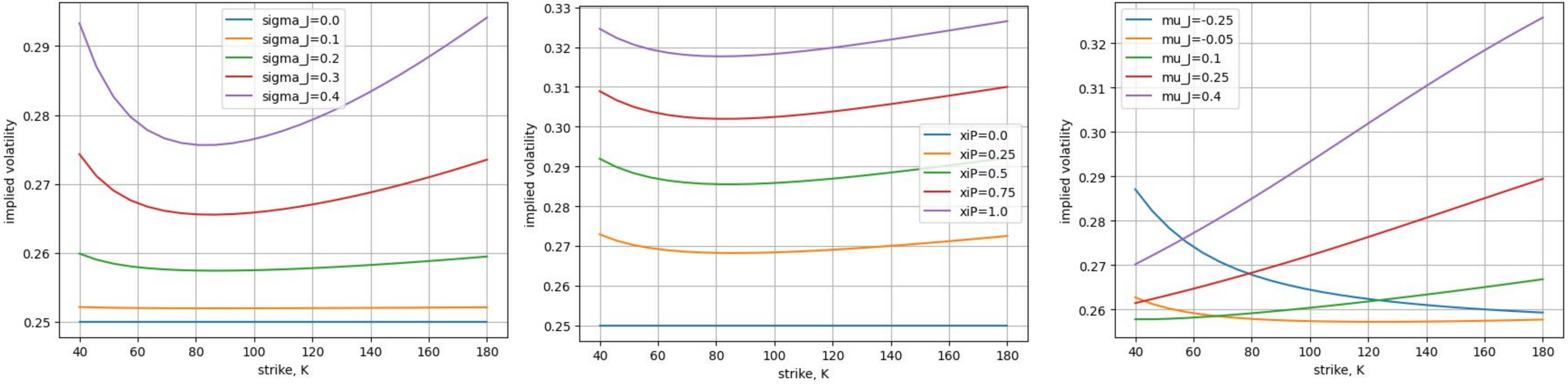
with boundary cases $Hh_{-1}(x) = e^{-x^2/2}$ (which equals $\sqrt{2\pi}\,\phi(x)$, where ϕ is the standard normal PDF) and $Hh_0(x) = \sqrt{2\pi}\,\Phi(-x)$.

The Hh_n functions satisfy the three-term recursion

$$n\,Hh_n(x) = Hh_{n-2}(x) - x\,Hh_{n-1}(x), \quad n\geq 1, \tag{41}$$

so all values for $n \geq 1$ can be computed from the normal density ϕ and normal CDF Φ alone.

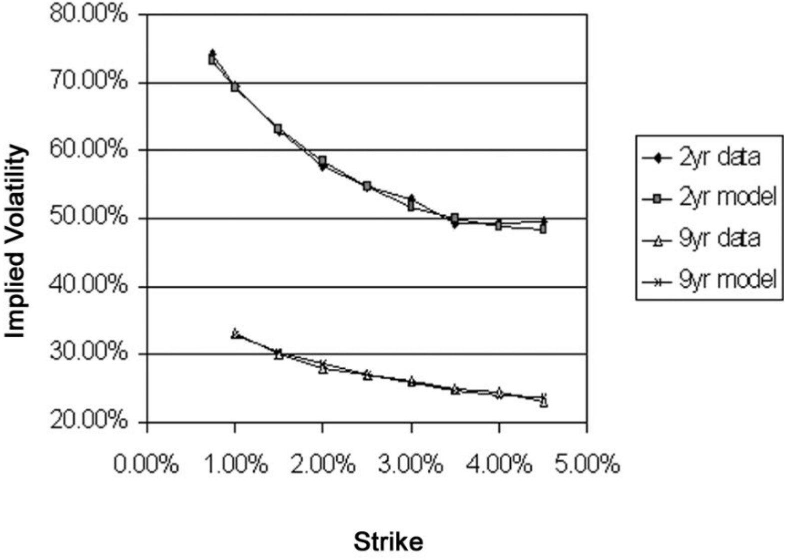
Application: Volatility Smile



- Merton’s Gaussian Jump Size is not capable to produce the smile, but Kou’s can
- Issue of Kou’s model: hard to tell the source of the smile

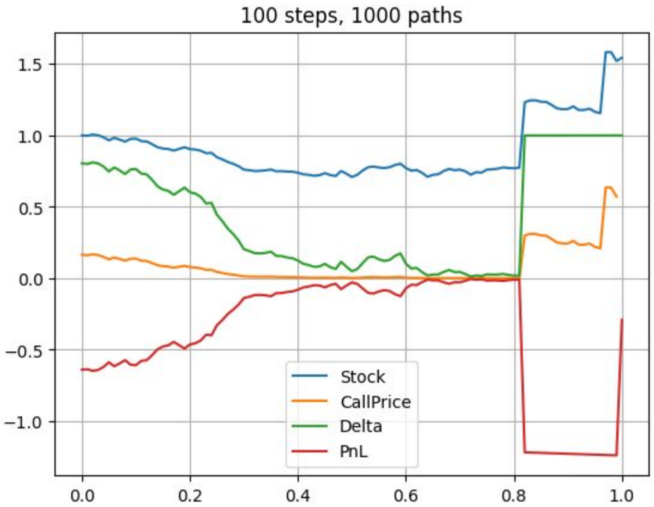
Empirical Volatility Smile

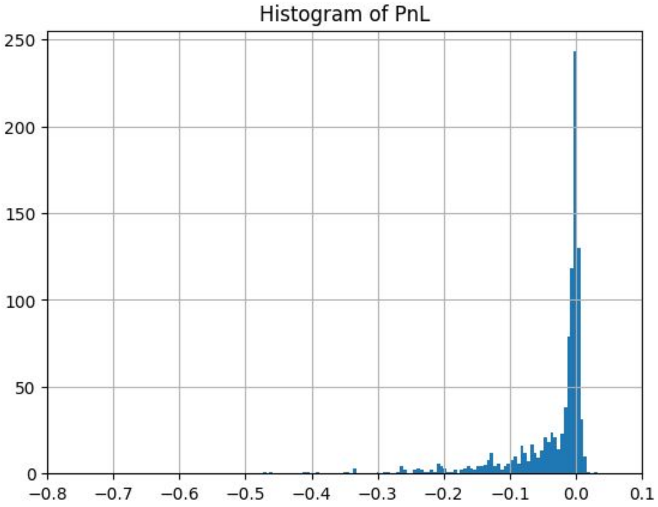
Figure 3 Midmarket and Model-Implied Volatilities for Japanese LIBOR Caplets in May 1998



Notes. The parameters used in the fitted model are: for the two-year caplet $\eta^{(1)} = 3.7$, $\eta^{(2)} = 1.8$, $p = 0.04$, $\lambda = 1.4$, $\sigma = 0.21$; and for the nine-year caplet $\eta^{(1)} = 2.3$, $\eta^{(2)} = 1.8$, $p = 0.09$, $\lambda = 0.2$, $\sigma = 0.09$.

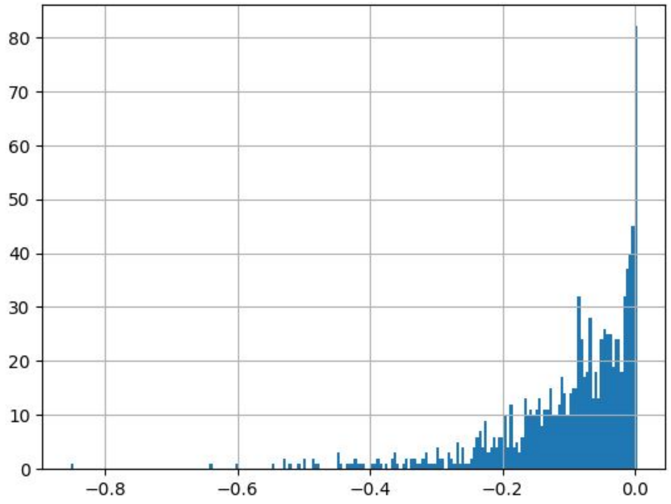
Delta Effect of a Jump





- The PnL distribution is skewed to have more weight in the negative region, with some heavy left tail

Delta Effect of Multiple Jump



- More steps => more jump => more skewed distribution
- The delta hedging in BS model can’t capture the jump <- loss realized

Theoretical Pricing Comparison

```
============================================================
Model                                       Call        Put
------------------------------------------------------------
Parameters:
S = 80, K = 81, T = 0.25, r = 0.08, sigma = 0.2
Lambda = 10, muJ = -0.02, sigmaJ = 0.02, N = 50
------------------------------------------------------------
Black-Scholes (no jumps)                   $3.49      $2.89
Merton jump-diffusion                      $3.78      $3.18
Kou jump-diffusion                         $3.92      $3.31
============================================================
```

- Jump only increase the volatility, never the other way around
- Option will be more expensive under Jump model

Issue of Jump Diffusion Model

1. Market Incompleteness: Jump is non-perfectly hedgeable, but the Risk-natural measure still exist, therefore there will be **no uniqle Q-pricing**
   - Researcher (e.g. Kou, Tsay, Grzelak) assumed **Jump risk can be diversified** with a portfolio of assets **sharing the same jump source**
2. Memoryless Jump <- unable to capture volatility clustering

Coupons / Multiple Debt Structure

1 Discrete Coupons in a Merton Model

The Merton model cannot price coupon-bearing debt by simply pricing each component separately. Pricing requires examining all coupon payments within a single integrated model, making assumptions about asset sales critical.

2 Two-Coupon Example

Consider a coupon bond with two coupons D_1 and D_2 payable at dates t_1 and t_2.

2.1 After First Coupon ($t > t_1$)

If the firm survives past t_1 with assets worth A_t, the remaining coupon D_2 can be valued using the standard Merton model:

$$B(A_t, t) = D_2 b(t, t_2) - P^{BS}(A_t, D_2, t_2 - t) \tag{1}$$

2.2 At First Coupon Date (t_1)

The situation at t_1 is more complex and depends critically on assumptions about equity owners' control over firm assets.

2.2.1 No Asset Sales Assumption

Equity owners cannot use firm assets to pay debt. They must finance debt payments either:

- Out of their own pockets, or
- By issuing new equity

Under no information asymmetries, both options are equivalent. If issuing M new shares to N existing shares:

$$\frac{M}{M+N} S_{t_1} = D_1 \tag{2}$$

where S_t is total equity value at time t.

This dilutes existing equity from S_{t_1} to $(N/(M+N))S_{t_1}$, causing a fall of $S_{t_1} - D_1$. Hence, paying out of pocket or dilution yield the same result.

Default condition: Equity owners pay coupon D_1 if and only if:

$$D_1 < C(A_{t_1}, D_2, t_2 - t_1) \tag{3}$$

Otherwise, default occurs and debtholders take over the firm.

Pricing Algorithm for Multiple Debt

If we know the structure of the debt/ coupon, we can iteratively compute the equity valuation at each payment.
If the **equity hit zero** in of the steps, we can conclude the corporate is unable to repay the existing debt structure, hence **default**.

3 Recursive Pricing Algorithm (No Asset Sales)

Given N coupons $D_1, \ldots, D_N$ at dates $t_1, \ldots, t_N$:

1. Price debt and equity at dates $t > t_{N-1}$ using standard Merton setup.
2. At t_{N-1}, find the critical value $\tilde{A}_{N-1}$ where:
$$D_{N-1} = C(\tilde{A}_{N-1}, t_N - t_{N-1}, D_N) \tag{4}$$
3. At date t_{N-1}, set boundary conditions:
$$S(A, t_{N-1}) = \begin{cases} C(A, D_N, t_N - t_{N-1}) - D_{N-1} & \text{for } A \geq \tilde{A}_{N-1} \\ 0 & \text{for } A < \tilde{A}_{N-1} \end{cases} \tag{5}$$
$$B(A, t_{N-1}) = \begin{cases} D_{N-1} + V - C(A, D_N, t_N - t_{N-1}) & \text{for } A \geq \tilde{A}_{N-1} \\ A & \text{for } A < \tilde{A}_{N-1} \end{cases} \tag{6}$$
4. Numerically value equity right after coupon payment at date t_{N-2}.
The value $\tilde{A}_{N-2}$ satisfies: equity is worth D_{N-2} right after the coupon payment at t_{N-2}.
5. Use the same procedure as in (3) to set boundary conditions at date t_{N-2} and continue recursively backwards to time 0.

Asset selling model for Multiple Coupon

4 Asset Sales Model

When asset sales are allowed, we still work recursively backwards but must adjust both the default boundary and asset value.

4.1 Optimal Strategy at t_{N-1}

If assets are worth more than D_{N-1}, default is never optimal (leaves equity with zero). Instead, equity owners should:

- Sell D_{N-1} worth of assets to cover the coupon
- Continue with assets worth $A(t_{N-1}) - D_{N-1}$
- Equity is then worth $C(A_{t_{N-1}} - D_{N-1}, D_N, t_N - t_{N-1})$

4.2 Comparison with Payment Out of Pocket

Paying out of pocket leaves equity with:

$$C(A_{t_{N-1}}, D_N, t_N - t_{N-1}) - D_{N-1} \tag{7}$$

But this is smaller than $C(A_{t_{N-1}} - D_{N-1}, D_N, t_N - t_{N-1})$ since $\delta_A C < 1$ for all V, giving:

$$C(A_{t_{N-1}}) - C(A_{t_{N-1}} - D_{N-1}) < D_{N-1} \tag{8}$$

This is intuitively obvious: coupon payment by equity owners alone benefits both equity and debt, but is an expense to equity only.

Takeaway: Equity owner prefers to sell asset over paying out of their pocket/diluting their shares.

Empirical Implementation Issue

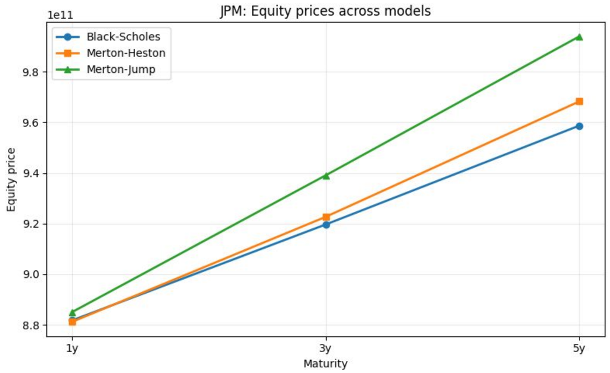
- **Asset process is unobservable** to public; Balance sheet released quarterly (as most) for public company
- One will need to use the fundamental accounting equation to approximate the asset process

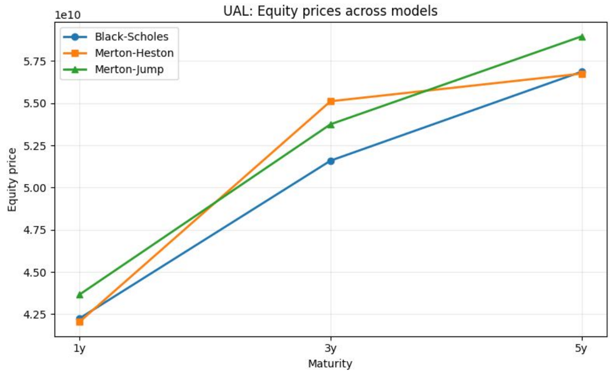
$$E_t = \text{Market Cap} = \#\text{Outstanding sharing} \times S_t$$

$$A_t = E_t - D$$

- Similarly, the **volatility of an asset process is hard to obtain**
- I used the realized stock volatility, another choice is the market traded implied volatility
- The idea is that the we aims to find the **relative valuation**, so it would be a fair pricing as long as the compared companies are **also using the same volatility scheme**

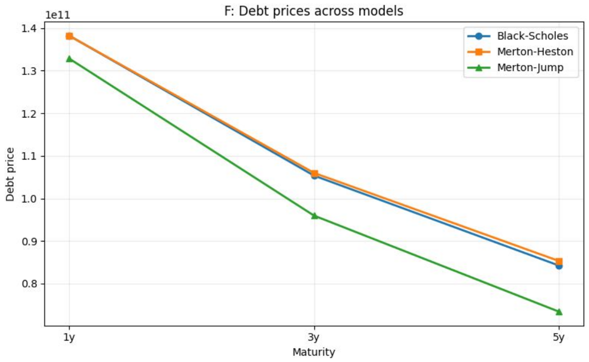
Equity Valuation Comparison

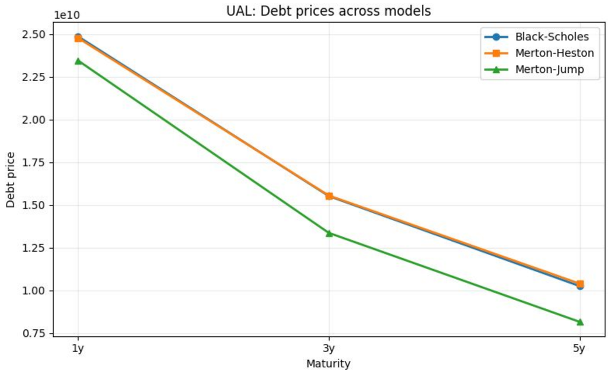




- Debt = Total Debt stated on the balance sheet
- Maturity: Set 1y, 3y, 5y (Actually debt is likely not following this assumption)

Debt Valuation Comparison





- Used call-put parity
- Jump model implies a more expensive put price, debt value is in short-put position thus its value is lower than continuous model

Default Probability & Distance to Default

Default Probability and Distance to Default

Structural default condition

In all models, default at maturity T occurs when the firm's asset value falls below the debt face value:

$$A_T < D.$$

Under the physical measure P, the physical default probability is

$$\text{PD}_{\text{physical}} = P(A_T < D),$$

and under the risk-neutral measure Q, the risk-neutral default probability is

$$\text{PD}_{\text{risk-neutral}} = Q(A_T < D).$$

The **Distance to Default (DD)** measures how many standard deviations the asset value is away from the default threshold.

Summary Table

Model	Jump distribution	DD formula	PD computation
Black–Scholes	None	Closed-form	$\Phi(-\text{DD})$
Merton Jump	$Y \sim \mathcal{N}(\mu_J, \sigma_J^2)$	$-\Phi^{-1}(\text{PD})$	Poisson sum (50 terms)
Kou	$Y \sim \text{Laplace}(\kappa, \eta)$	$-\Phi^{-1}(\text{PD})$	Fourier (Gil–Pelaez)

All implementations follow these formulas directly.

Default Probability & Distance to Default

Model 1: Black–Scholes (No Jumps)

Asset dynamics

Under the physical measure:

$$\frac{dA_t}{A_t} = \mu\, dt + \sigma_A\, dW_t^P,$$

so

$$\log A_T \sim \mathcal{N}\left(\log A_0 + \left(\mu - \tfrac{1}{2}\sigma_A^2\right) T,\ \sigma_A^2 T\right).$$

Distance to Default

$$\mathrm{DD}_{\mathrm{BS}} = \frac{\log(A_0/D) + (\mu - \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}.$$

Default probabilities

Physical measure:

$$\mathrm{PD}^{\mathrm{BS}}_{\text{physical}} = P(A_T < D) = P(\log A_T < \log D) = \Phi(-\mathrm{DD}_{\mathrm{BS}}).$$

Risk-neutral measure (replace μ with r):

$$\mathrm{PD}^{\mathrm{BS}}_{\text{risk-neutral}} = \Phi\left(-d_2^Q\right), \quad d_2^Q = \frac{\log(A_0/D) + (r - \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}.$$

Default Probability & Distance to Default

ticker	maturity_years	Distance_to_Default_Black-Scholes	Distance_to_Default_Merton_Jump	Distance_to_Default_Kou
AAPL	1.0	11.660	7.330	5.969
AMZN	1.0	8.247	5.691	6.016
F	1.0	1.703	1.276	1.647
JPM	1.0	5.172	3.202	4.837
KO	1.0	12.378	5.282	6.059
T	1.0	3.860	2.299	3.575
UAL	1.0	1.313	1.149	1.295
XOM	1.0	11.308	5.969	6.025

Default Probability & Distance to Default

ticker	PD_physical_Black-Scholes	PD_physical_Merton_Jump	PD_physical_Kou	PD_risk_neutral_Black-Scholes	PD_risk_neutral_Merton_Jump	PD_risk_neutral_Kou
AAPL	1.025e-31	1.154e-13	1.191e-09	3.003e-31	9.426e-14	1.182e-09
AMZN	8.091e-17	6.322e-09	8.958e-10	7.055e-17	4.117e-09	8.976e-10
F	4.432e-02	1.010e-01	4.982e-02	2.174e-01	2.754e-01	2.260e-01
JPM	1.157e-07	6.835e-04	6.602e-07	2.342e-05	4.230e-03	6.496e-05
KO	1.728e-35	6.405e-08	6.854e-10	2.670e-32	1.290e-07	6.519e-10
T	5.669e-05	1.075e-02	1.751e-04	2.763e-04	1.672e-02	6.735e-04
UAL	9.458e-02	1.254e-01	9.763e-02	1.205e-01	1.484e-01	1.239e-01
XOM	6.002e-30	1.191e-09	8.459e-10	1.908e-27	2.869e-09	8.072e-10

PD & DD Appendix - Merton's Model

Model 2: Merton Jump-Diffusion

Asset dynamics with jumps

Under the physical measure:

$$\frac{dA_t}{A_t} = (\mu - \lambda k)\, dt + \sigma_A \, dW_t^P + d\left(\sum_{i=1}^{N_t}(J_i - 1)\right),$$

where $N_t \sim \text{Poisson}(\lambda t)$, $J_i = e^{Y_i}$ with $Y_i \sim \mathcal{N}(\mu_J, \sigma_J^2)$ i.i.d., and

$$k = E[J_i - 1] = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1.$$

Conditional lognormal distribution

Conditional on $N_T = n$ jumps, the asset value is lognormal:

$$A_T \mid N_T = n \sim \text{Lognormal}\big(\log A_0 + \big(r_n - \tfrac{1}{2}\sigma_n^2\big)\, T,\ \sigma_n^2 T\big),$$

with adjusted parameters

$$r_n = \mu - \lambda k + \frac{n\gamma}{T}, \quad \gamma = \log(1+k) = \mu_J + \tfrac{1}{2}\sigma_J^2,$$

$$\sigma_n^2 = \sigma_A^2 + \frac{n\sigma_J^2}{T}.$$

Default probabilities via Poisson mixture

Physical measure: The jump count follows $N_T \sim \text{Poisson}(\lambda T)$. Averaging over all possible jump counts:

$$\text{PD}_{\text{physical}}^{\text{Merton}} = \sum_{n=0}^{\infty} \frac{e^{-\lambda T}(\lambda T)^n}{n!}\, \Phi\left(-\frac{\log(A_0/D) + (r_n^P - \frac{1}{2}\sigma_n^2)T}{\sigma_n \sqrt{T}}\right),$$

where $r_n^P = \mu - \lambda k + n\gamma/T$.

Risk-neutral measure: Under Q, the jump intensity shifts to $\lambda' = \lambda(1+k)$ and the drift becomes r:

$$\text{PD}_{\text{risk-neutral}}^{\text{Merton}} = \sum_{n=0}^{\infty} \frac{e^{-\lambda' T}(\lambda' T)^n}{n!}\, \Phi\left(-\frac{\log(A_0/D) + (r_n^Q - \frac{1}{2}\sigma_n^2)T}{\sigma_n \sqrt{T}}\right),$$

where $r_n^Q = r - \lambda k + n\gamma/T$.

Distance to Default (pseudo-DD)

Since the distribution is no longer exactly lognormal, we define an *equivalent* Distance to Default:

$$\text{DD}_{\text{Merton}} = -\Phi^{-1}\big(\text{PD}_{\text{physical}}^{\text{Merton}}\big).$$

This is the standard-normal distance such that $\Phi(-\text{DD}_{\text{Merton}}) = \text{PD}_{\text{physical}}^{\text{Merton}}$. When $\lambda = 0$ (no jumps), $\text{DD}_{\text{Merton}} = \text{DD}_{\text{BS}}$ exactly.

Note: In practice, the infinite series is truncated at $N_{\max} = 50$ terms.

PD & DD Appendix - Kou's Model

Model 3: Kou Double-Exponential Jump-Diffusion

Asset dynamics with Laplace jumps

Under the physical measure:

$$\frac{dA_t}{A_t} = (\mu - \lambda\psi)\, dt + \sigma_A\, dW_t^P + d\left(\sum_{i=1}^{N_t}(J_i - 1)\right),$$

where $N_t \sim \text{Poisson}(\lambda t)$, $Y_i = \log J_i \sim \text{Laplace}(\kappa, \eta)$ i.i.d., and

$$\psi = E[J_i - 1] = \frac{e^{\kappa}}{1-\eta^2} - 1, \quad 0 < \eta < 1.$$

The Laplace (double-exponential) density for Y_i is

$$f_Y(y) = \frac{1}{2\eta}\, e^{-|y-\kappa|/\eta}.$$

Characteristic function

The log-return $X_T = \log(A_T/A_0)$ has characteristic function

$$\varphi_{X_T}(u) = \exp\!\left(iu\,\mu_x\,T - \tfrac{1}{2}\sigma_A^2\,u^2\,T + \lambda T\,(\varphi_Y(u) - 1)\right),$$

where

$$\mu_x = \mu - \tfrac{1}{2}\sigma_A^2 - \lambda\psi, \quad \varphi_Y(u) = \frac{e^{iu\kappa}}{1+\eta^2 u^2}.$$

Default probabilities via Gil–Pelaez inversion

The CDF of X_T at threshold $\log(D/A_0)$ is computed via Fourier inversion:

$$P(X_T < a) = \frac{1}{2} - \frac{1}{\pi}\int_0^{\infty} \frac{\text{Im}\!\left[e^{-iua}\,\varphi_{X_T}(u)\right]}{u}\, du.$$

Physical measure:

$$\text{PD}^{\text{Kou}}_{\text{physical}} = P(\log A_T < \log D) = P\left(X_T < \log\left(\frac{D}{A_0}\right)\right),$$

computed numerically using adaptive quadrature with $\mu_x = \mu - \frac{1}{2}\sigma_A^2 - \lambda\psi$.

Risk-neutral measure (replace μ with r):

$$\text{PD}^{\text{Kou}}_{\text{risk-neutral}} = Q\left(X_T < \log\left(\frac{D}{A_0}\right)\right),$$

with $\mu_x^Q = r - \frac{1}{2}\sigma_A^2 - \lambda\psi$.

Distance to Default (pseudo-DD)

As with the Merton Jump model, we define

$$\text{DD}_{\text{Kou}} = -\Phi^{-1}\left(\text{PD}^{\text{Kou}}_{\text{physical}}\right).$$

When $\lambda = 0$, $\text{DD}_{\text{Kou}} = \text{DD}_{\text{BS}}$ exactly.

Conclusion

Next step:

- Expand the empirical result (e.g. small cap, junk bond, government treasury vs corporate bond)
- Compute the Expected Loss = PD * LGD (Loss given Default) * EAD (Exposure at Default)
- Credit Spread
- Beyond Jump Model (Black-Cox model, Copulas model)
- Jump hedging scheme (Portfolio Diversification Hedging)

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Appendix - Credit Spread

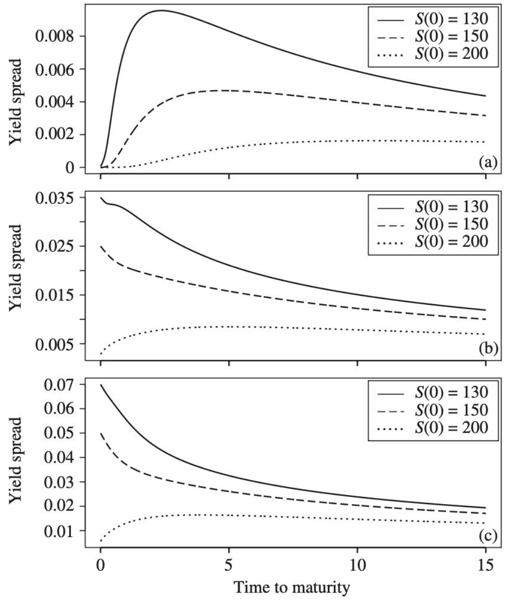

Figure 2.7. The effect of changing the mean jump size and the intensity in a Merton model with jumps in asset value. From (a) to (b) we are changing the parameter determining the mean jump size, γ, from log(0.9) to log(0.5). This makes recovery at an immediate default lower and hence increases the spread in the short end. From (b) to (c) the intensity is doubled, and we notice the exact doubling of spreads in the short end, since expected recovery is held constant but the default probability over the next short time interval is doubled.

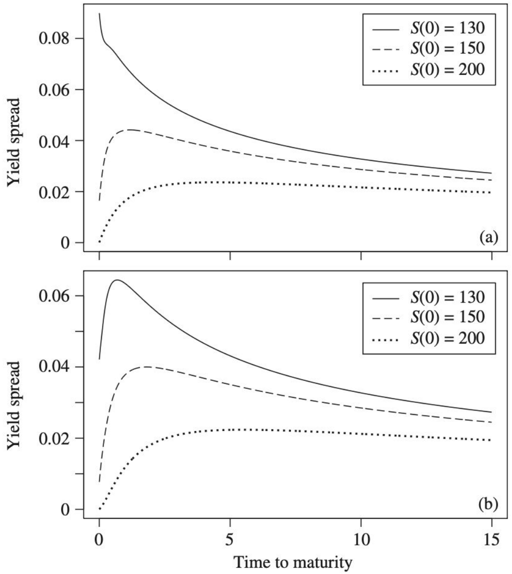

Figure 2.8. The effect of the source of volatility in a Merton model with jumps in asset value. (a) The diffusion part has volatility $\sigma = 0.1$, and the total quadratic variation is 0.4. (b) The diffusion part has volatility $\sigma = 0.3$, but the total quadratic variation is kept at 0.4 by decreasing λ. In both cases, three different current asset values are considered. Changing the source of volatility causes significant changes of the yield spreads in the short end for the high-yield cases. The difference between the spread curves in the case of low leverage is very small. Effects are also limited in the long end in all cases.

Appendix - SemiMartingales Itô's Lemma

D.4 The General Itô Formula for Semimartingales

In this section we first present a version of Itô's formula which is general enough to handle the processes encountered in this book. Assume throughout that the semimartingale X satisfies the following property:

$$\sum_{0 \leqslant s \leqslant t} |\Delta X_s| < \infty.$$

Certainly, a process which only jumps a finite number of times on any finite time interval satisfies this property. With this property satisfied, we can define the continuous part of the process X as

$$X_t^c = X_t - \sum_{0 \leqslant s \leqslant t} \Delta X_s.$$

This continuous part of X also has a continuous martingale part, and we refer to the quadratic variation of this as $\langle X^c \rangle$. We have not formally defined the predictable quadratic variation but we will do that below and will write it out for the case of a jump-diffusion.

Let f be a function depending on x and t which is C^2 in the x variable and C^1 in the t variable. Then $f(X_t, t)$ is also a semimartingale and

$$\begin{aligned} &f(X_t) - f(X_0) \\ &\quad = \int_0^t f'(X_s)\,\mathrm{d}X_s^c + \int_0^t \tfrac{1}{2} f''(X_s)\,\mathrm{d}\langle X^c \rangle_s + \sum_{0 \leqslant s \leqslant t} \{f(X_s, s) - f(X_{s-}, s)\}. \end{aligned}$$